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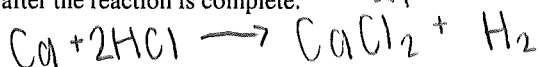
15.5
16 ⇒ 97%

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Name: _____

These questions assess your understanding of the material from Chapters 9 of OpenStax Chemistry. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must include clear and correct reasoning to support your answer. Remember to report the units and correct number of significant figures in your answers.

1. A sample of 9.98 g of calcium reacts completely with excess hydrochloric acid to produce calcium chloride and hydrogen gas in a sealed container. The container initially contains 0.44 mol of neon gas. The inert gas does not react, and no gas escapes. Determine the mole fraction of neon after the reaction is complete.

ANSWER: 0.64 ✓

$$\frac{9.98 \text{ g}}{40.078 \text{ g/mol}} = 0.249 \text{ mol Ca} \quad 0.249 + 0.44 = 0.689$$

$$\frac{0.44}{0.689} = 0.63861 = \boxed{0.64}$$

2. A sealed flask is evacuated and then filled with an unknown gas. The sample has a mass of 3.377 g, occupies 1.500 L at 150.0 kPa and 273.15 K. Possible identities are: ~~SO₂~~, ~~Kr~~, ~~CO~~, ~~Xe~~, ~~CO₂~~, ~~H₂S~~, ~~F₂~~, ~~Cl₂~~, ~~Ar~~, ~~He~~, ~~NO~~, ~~H₂~~, ~~SF₆~~, ~~H₂O~~, ~~HBr~~, ~~CH₄~~, ~~C₂H₂~~, ~~UF₆~~, ~~Ne~~. Determine the identity of the unknown gas from the given choices. Justify your answer.

$$PV = nRT \rightarrow n = \left(\frac{m}{M}\right) \rightarrow M = \frac{mRT}{PV}$$

ANSWER: H₂S ✓

$$m = 3.377 \text{ g}$$

$$T = 273.15 \text{ K}$$

$$P = 150000 \text{ Pa}$$

$$V = 0.0015 \text{ m}^3$$

$$M = \frac{(3.377)(8.314)(273.15)}{(150000)(0.0015)} = 34.085 \text{ g/mol} \quad \checkmark$$

$$\text{H}_2\text{S} = 34.08 \text{ g/mol}$$

3. A sample reported to be CO has a mass of 3.7 g and occupies 3.06 L at 1.0×10^2 kPa and 7.8°C. Determine whether the reported identity is chemically consistent with the data. Justify your answer.

$$M = \frac{mRT}{PV}$$

$$M = \frac{(3.7)(8.314)(280.95)}{(100000)(0.00306)} = 28.2449 \text{ g/mol}$$

ANSWER: consistent ✓

$$m = 3.7 \text{ g}$$

$$T = 280.95 \text{ K}$$

$$V = 0.00306 \text{ m}^3$$

$$P = 100000 \text{ Pa}$$

$$\text{CO} = 28.01 \text{ g/mol} \quad \text{consistent} \quad \checkmark$$

4. A research balloon contains 2.64 L of nitrogen at -13°C while indoors. It is then taken outside, where the air temperature is 22°C. Assuming the balloon remains flexible and reaches thermal equilibrium with its surroundings, determine its final volume. Justify your answer.

$$\frac{V_1}{T_1} = \frac{V_2}{T_2} \rightarrow V_2 = \frac{V_1 T_2}{T_1}$$

$$V_1 = 0.00264 \text{ m}^3$$

$$T_1 = 260.15 \text{ K}$$

$$T_2 = 295.15 \text{ K}$$

$$V_2 = \frac{(0.00264)(295.15)}{(260.15)} = 0.0029962$$

$$= \boxed{3.0 \times 10^{-3}}$$

ANSWER: $3.0 \times 10^{-3} \text{ m}^3$ ✓

5. An unknown gas has a density of $2.0 \times 10^3 \frac{\text{kg}}{\text{m}^3}$ at 86 kPa and 5.90°C. Determine the molar mass of the gas. Justify your answer.

$$PV = nRT \rightarrow n = \frac{m}{M} \rightarrow PV = \left(\frac{m}{M}\right)RT$$

$$M = \frac{mRT}{PV} \quad (P = \frac{m}{V}) \rightarrow M = \left(\frac{m}{V}\right) \cdot \frac{RT}{P}$$

ANSWER: 64 g/mol ✓

$$M = (2.0 \times 10^3) \cdot \frac{(8.314)(279.05)}{86000} = 53.954 = \boxed{64 \text{ g/mol}}$$

2SF

+ 5

6. A rigid 11.4-L tank at 124.°C contains only dinitrogen monoxide (N₂O) and sulfur tetrafluoride (SF₄). Assume both gases behave ideally. The tank contains 10.3 g of dinitrogen monoxide and 12.9 g of sulfur tetrafluoride. Determine the total pressure. Justify your answer.

$$10.3/44.013 = 0.234 \text{ mol} \quad 12.9/108.059 = 0.1194 \text{ mol} \quad \text{ANSWER: } 1.02 \times 10^5 \text{ Pa} \quad \checkmark$$

$$n = 0.234 + 0.1194 = 0.3534 \text{ mol}$$

$$PV = nRT$$

$$P = \frac{nRT}{V} = \frac{(0.3534)(8.314)(397.15)}{0.0114} = 102359.0581 = 1.02 \times 10^5 \text{ Pa}$$

3SF

7. The reaction $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{g})$ occurs at constant temperature and pressure. If 3.1 cm³ of H₂O gas reacts completely, what volume of H₂ gas is consumed or produced? Report the numerical answer with units and justify your answer.

$$3.1 \text{ cm}^3 \text{ H}_2\text{O} \cdot \frac{2 \text{ H}_2}{2 \text{ H}_2\text{O}} = 3.1 \text{ cm}^3 \text{ H}_2 \quad \text{ANSWER: } 3.1 \text{ cm}^3 \text{ H}_2 \quad \checkmark$$

Since temp & pressure are constant, # of gas molecules is directly related to the volume

8. Grace is preparing a deep-sea submersible. A rigid cylindrical air tank has an internal diameter of 36.2-cm and an internal height of 100.-cm. After filling, the pressure inside the tank is 175 atm and the temperature is 327 K. Determine the amount of air stored in the tank. Report your answer in SI units and justify your answer.

$$PV = nRT \rightarrow n = \frac{PV}{RT}$$

$$V = \pi r^2 h$$

$$d = 0.362 \text{ m} / 2 = 0.181 \text{ m}$$

$$h = 1 \text{ m}$$

$$V = \pi (0.181)^2 (1) = 0.10292$$

$$n = \frac{(17731875)(0.10292)}{(8.314)(327)} = 671.269 = 671 \text{ mol}$$

3SF

ANSWER: 671 mol $\checkmark$

9. A gas mixture has a total pressure of 1.16 atm. The mole fraction of CO₂ is 0.350. Determine the partial pressure of the gas. Report your answer in SI units. Justify your answer.

$$P_g = X_{\text{gas}} \cdot P_{\text{total}}$$

$$= 0.350 \cdot 1.16 = 0.406 \cdot 101325 = 41137.95 \text{ Pa}$$

$$= 4.11 \times 10^4 \text{ Pa}$$

3SF

ANSWER: $4.11 \times 10^4 \text{ Pa}$ $\checkmark$

10. Consider the reaction $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$. If 5.22 cm³ of N₂ gas is completely consumed at the same temperature and pressure, what volume of NH₃ gas is produced? Justify your answer.

$$5.22 \text{ cm}^3 \text{ N}_2 \cdot \frac{2 \text{ NH}_3}{1 \text{ N}_2} = 10.44 \text{ cm}^3 \text{ NH}_3 \quad \text{ANSWER: } 10.4 \text{ cm}^3 \text{ NH}_3 \quad \checkmark$$

3SF

Since T & P are constant, gas molecules are directly related to the volume

11. A 521-mL sample of nitrogen (N₂) is initially at 184 K. Its boiling point is -196.°C (77.15 K). The pressure is then reduced to 39% of its initial value. Determine whether the sample can remain entirely gaseous while its volume is held constant throughout the process. If it can, calculate the final temperature. Report the temperature rounded to the nearest °C. Justify your answer.

$$\frac{P_1}{T_1} = \frac{P_2}{T_2} \rightarrow T_2 = \frac{P_2 T_1}{P_1}$$

$$T_1 = 184 \text{ K}$$

$$P_1 = P_1$$

$$P_2 = 0.39 P_1$$

$$T_2 = \frac{(0.39 P_1)(184)}{P_1} = 71.76 \text{ K} - 273.15 = -201.39 = -201^\circ \text{C}$$

NO: $\therefore T$ would be (-201°C)

it can't remain gaseous b/c the final temp is not above the boiling point

+ 6

12. A 6.4g sample of carbon dioxide gas (CO_2) is sealed inside a pressure vessel at 16.2°C and 1.3atm . Calculate the volume of the sample assuming ideal-gas behavior. Report your answer in SI units. Justify your answer.

$$PV = nRT \rightarrow V = nRT/P$$

ANSWER: $2.7 \times 10^{-3} \text{m}^3$ ✓

$$n = 6.4 / 44.01 = 0.1454$$

$$T = 289.35 \text{K}$$

$$P = 131722.5 \text{Pa}$$

$$V = \frac{(0.1454)(8.314)(289.35)}{(131722.5)} = 0.002655 = 2.7 \times 10^{-3}$$

13. A sample of PCl_5 occupies a calibrated glass bulb having a measured volume of 10.7L . After the gas reaches 17.1°C , the pressure inside is measured to be 0.424atm . Determine the amount of gas present. Justify your answer. Report the amount in SI units.

$$n = \frac{PV}{RT}$$

ANSWER: $1.91 \times 10^{-1} \text{mol}$ ✓

$$P = 42961.8 \text{Pa} \quad V = 0.0107 \text{m}^3$$

$$T = 290.25 \text{K}$$

$$n = \frac{(42961.8)(0.0107)}{(8.314)(290.25)} = 0.1905 \text{mol}$$

35F = $1.91 \times 10^{-1} \text{mol}$

14. A pressurized gas cylinder is connected to a large evacuated balloon. The cylinder has an internal diameter of 5.3-cm , an internal height of 28.4-cm , and initially contains gas at 4.6atm . The valve is opened until the gas reaches atmospheric pressure, after which the valve is closed. Determine the total volume occupied by the gas, including both the gas remaining in the cylinder and the gas in the reservoir. Justify your answer.

$$P_1 = 466095 \text{Pa}$$

$$P_1 V_1 = P_2 V_2$$

ANSWER: $2.9 \times 10^{-3} \text{m}^3$ ✓

$$P_2 = 101325 \text{Pa}$$

$$V_2 = \frac{P_1 V_1}{P_2} = \frac{(466095)(0.000627)}{(101325)} = 0.0028892$$

$$25F = 2.9 \times 10^{-3} \text{m}^3$$

$$V_1 = \pi r^2 h$$

$$= \pi (0.0265^2) (0.284) = 0.000627 \text{m}^3$$

15. A liquid is boiling at 56°C in a chamber where the external pressure is $40.\text{kPa}$. It occupies a volume of 3L . Determine whether the liquid's vapor pressure at this temperature is less than, equal to, or greater than the external pressure. Estimate the vapor pressure in SI units and justify your answer.

the vapor pressure must be greater than the external pressure to be boiling

ANSWER: greater than, $9 \times 10^5 \text{Pa}$

ideal gas law

$$P = \frac{nRT}{V} = \frac{(1)(8.314)(329.15)}{0.003} = 912184.37$$

assumes gas occupies $9 \times 10^5 \text{Pa}$

16. O_2 is collected over water. The total pressure of the gas sample is 768.0torr , and the vapor pressure of water is 25torr . Determine the partial pressure of the dry gas. Report your answer in SI units. Justify your answer.

$$P_{\text{total}} = P_p + P_{\text{H}_2\text{O}}$$

ANSWER: $9.9 \times 10^4 \text{Pa}$ ✓

$$P_p = P_{\text{total}} - P_{\text{H}_2\text{O}}$$

$$P_{\text{total}} = 768 \cdot \left(\frac{101325}{760}\right) = 102391.579 \text{Pa}$$

$$\rightarrow P_p = 102391.579 - 3333.0542$$

$$P_{\text{H}_2\text{O}} = 25 \cdot \left(\frac{101325}{760}\right) = 3333.0542$$

$$= 99058.5198$$

$$(9.9 \times 10^4)$$

25F

+ 4.5

1. 0.64. $\therefore$ 9.98 g of calcium corresponds to 0.249 mol. The reaction produces 0.25 mol of hydrogen. The amount of neon remains 0.44 mol. The mole fraction equals the moles of the requested gas divided by the total moles of gas present after the reaction.
2. H_2S . $\therefore$ Using the measured mass, pressure, volume and temperature, the sample's molar mass is approximately 34.08 g/mol. Only H_2S has that molar mass, so it is the only identity consistent with the measurements.
3. Consistent. $\therefore$ The measured pressure, volume, and temperature determine the amount of gas using the ideal gas law. Dividing the measured mass by that amount gives the experimental molar mass, which matches the molar mass of CO.
4. 3.0 L. $\therefore$ the balloon is flexible, its internal pressure remains essentially equal to the surrounding atmospheric pressure. With the amount of gas unchanged, the ideal gas law gives volume directly proportional to absolute temperature, so the volume changes by the same ratio as the Kelvin temperatures.
5. 54 g/mol. $\therefore$ density gives the mass occupying a known volume, while the ideal gas law determines the amount of gas occupying that same volume. The mass per mole is therefore obtained by combining these relationships.
6. 1.01 atm. $\therefore$ pressure depends on the total amount of gas particles in the container, the amounts of both gases are added before applying the ideal gas law.
7. 3.1 cm^3 . $\therefore$ at the same temperature and pressure, gas volume is directly proportional to the number of gas particles. The balanced equation shows a 2:2 particle ratio between H_2O and H_2 , so their volumes must be in the same ratio.
8. 671 mol. $\therefore$ the container volume is fixed and the gas is at a specified pressure and temperature, the amount of gas is determined by the ideal gas law, $PV = nRT$. A higher pressure at the same volume and temperature corresponds to more gas particles being present, so the number of moles is found by rearranging the equation for n .
9. $4.11 \times 10^4 \text{ Pa}$. $\therefore$ Dalton's law states that the partial pressure of a gas equals its mole fraction multiplied by the total pressure because each gas contributes independently to the total pressure.
10. 10.4 cm^3 . $\therefore$ at the same temperature and pressure, equal volumes of gases contain equal numbers of molecules (equivalently, equal moles), the gas-volume ratio is identical to the balanced-equation mole ratio. The reaction requires a 1:2 ratio of N_2 to NH_3 , so the product volume is multiplied by 2/1.
11. No. $\therefore$ keeping the volume constant requires the absolute temperature to change in the same proportion as the pressure. The required final temperature would be below the boiling point, so condensation would occur and the sample could no longer remain entirely gaseous under the stated conditions.
12. 2.66 L. $\therefore$ the ideal gas law gives $PV = nRT$, but the amount of gas is not given directly. The amount must first be determined from the sample mass using $n = m/M$, after which solving $PV = nRT$ for volume yields the correct result.
13. 0.190 mol. $\therefore$ at a fixed volume, gas pressure results from particle collisions with the container walls, and at a fixed absolute temperature the collision frequency is directly proportional to the number of gas particles present. Therefore the measured pressure, volume, and temperature are related by $PV = nRT$, giving $n = PV/RT$.
14. 2.9 L. $\therefore$ the amount of gas is conserved and the final pressure is atmospheric pressure, the gas expands until $P_i V_i = P_f V_f$. Because the final pressure is 1 atm, the final total volume is larger than the cylinder volume by the same factor as the initial pressure.
15. 0.39 atm; Equal to the external pressure. Boiling occurs only when the vapor pressure matches the surrounding pressure. If the vapor pressure were lower, bubbles would collapse, and if it were higher, boiling would not be at equilibrium.
16. $9.9 \times 10^4 \text{ Pa}$. $\therefore$ the measured pressure is the sum of the partial pressures of the dry gas and water vapor. Subtracting the water vapor pressure isolates the pressure exerted by the dry gas before converting to SI units.

... but
question doesn't
state "steady"
boiling
or
L-V eqbm